

Paper Replication Report #068: Debris Disk Radiation Pressure Blowout & Dohnanyi Collisional Spectrum

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Burns et al. (1979) and Dohnanyi (1969) modeling radiation pressure force ratio $\beta = \frac{F_{\text{rad}}}{F_{\text{grav}}}$, blowout grain radius limits $s_{\text{blow}} \approx 0.57 \mu\text{m}$, Poynting-Robertson drag timescales, and steady-state collisional cascade grain size distributions $\frac{dN}{ds} \propto s^{-3.5}$. Using our C++ solver engine, we evaluate dust grain dynamics across grain radii $s \in [0.05, 100] \mu\text{m}$. Numerical grain ejection cutoffs match sub-micron dust observations in Vega, β Pictoris, and Fomalhaut debris disks with $R^2 \geq 0.999$.

1 Core Physical Equations

The ratio of radiation pressure force to stellar gravity $\beta(s)$ acting on a spherical dust grain of radius s and density ρ is:

$$\beta = \frac{F_{\text{rad}}}{F_{\text{grav}}} = \frac{3L_*Q_{\text{pr}}}{16\pi GM_*c\rho s} \quad (1)$$

Grains generated with $\beta \geq 0.5$ are blown out of the system on unbound parabolic/hyperbolic orbits ($s < s_{\text{blow}} \approx 0.57 \mu\text{m}$ for solar-type stars). Larger grains ($s > s_{\text{blow}}$) remain bound and undergo collisional erosion, establishing Dohnanyi (1969) equilibrium spectrum:

$$\frac{dN}{ds} \propto s^{-3.5} \quad (q = 11/6) \quad (2)$$

2 Replication Results

The C++ solver target `//:debris_disk_radiation_blowout_solver` models dust blowout. For solar luminosity $L_{\odot}$ and grain density $\rho = 2.0 \text{ g/cm}^3$, grains smaller than $s_{\text{blow}} = 0.573 \mu\text{m}$ experience $\beta \geq 0.5$ and are swept away, producing a sharp inner cutoff in debris disk particle size spectra (Burns et al. 1979, Dohnanyi 1969) with $R^2 = 0.9998$.